Lead Quality Analysis

RZR Analytics Challenge: trends, drivers and a plan to lift lead quality by 20%

Scope. 3,021 leads sold to one advertiser between 2009-04-02 and 2009-09-30, 24 data columns. Lead quality is measured as the share of leads that end **Closed** (became a customer): 245 of 3,021 = **8.1%**, matching the 8.0% quoted in the brief. All tests are two-sided at the 5% level; confidence intervals are 95% Wilson intervals.

Executive summary

Q1 Are there lead-quality trends over time?

Quality is drifting down and the drop is statistically significant, but it is fragile and partly a maturation effect.

• Closed rate fell from **9.1%** (Apr-Jun, n=1,707) to **6.8%** (Jul-Sep, n=1,314); difference p = 0.018. Logistic trend: odds fall 11% per 30 days (CI 0.82-0.97, p = 0.006); the decline survives adjustment for traffic mix (p = 0.001).

• But it is not significant once September is excluded (p = 0.18). September leads have had the least time to close: their share sitting at the pre-close 'EP Confirmed' stage is 10.8% vs 3-5% in other months, and the combined Closed + EP-stage rate is flat (p = 0.88).

• Read-out: a real decline is plausible but not proven. Re-run when September has matured.

Q2 What drives lead quality?

The traffic theme (ad group / keyword intent) is the dominant driver; debt level is second. The creative and form design are not.

• Leads from **student-loan / loan-default keywords** (477 leads, 15.8% of volume) closed at **1.0%** (5 closes) vs **9.4%** for everything else (p <0.001). They are 16% of volume but 2% of closes.

• The 'Holding Tank' ad group closes at **14.5%** (n=338), significantly above average after multiple-test correction. Debt level matters (p = 0.006): $70-100k is strong; under $10k and over $100k are weak.

• No reliable difference for widget design (p = 0.42), 1- vs 2-page form (p = 0.74), branded vs generic ad (p = 0.60), state, or placement.

Q3 Can we reach 9.6% (a 20% lift) and earn the +20% CPL?

Yes, there is a credible path. Most of the lift comes from one change; the last step needs better replacement traffic.

• Stopping student-loan traffic alone lifts quality from 8.1% to **9.4%** (+16%) while keeping 84% of volume. Its effect holds out-of-sample: Jul-Sep rate 6.8% to 8.0%.

• Reaching 9.6% at full volume needs the freed 477 leads replaced by traffic that closes at **at least 10.5%**. The best ad groups ('Holding Tank' + 'Consolidate') close at 13.8%, so this is feasible. Gross revenue would then be about **+20%** vs today.

• Without replacement, a $100k+ debt screen on top reaches 9.7% but volume falls to 80%, so revenue would be about 4% lower unless saved traffic cost offsets it.

Bottom line and what I would do first

Fix the traffic mix before touching creative.

• **Now:** stop student-loan / loan-default traffic (about 1% close rate, 16% of volume).

• **Next:** move the freed budget to the 12-15% ad groups ('Holding Tank', 'Consolidate', 'Stop Collections') and screen debt levels under $10k and over $100k.

• **Before agreeing the CPL deal:** fix the measurement window and metric with the advertiser (a 20% lift is hard to detect with about 500 leads a month), and re-run the trend on mature cohorts.

1. Data, definitions and checks

The dataset has one row per lead. The assignment brief defines four quality groups; I use them as follows.

Group	CallStatus values	Leads	Share
Closed (best)	Closed	245	8.1%
Good, not yet closed	EP Sent, EP Received, EP Confirmed	148	4.9%
Bad	Unable to contact, Invalid profile, Doesn't qualify	488	16.2%
Unknown	Blank CallStatus (treated as 'Unknown')	2140	70.8%

Primary metric: Closed % of all leads (denominator = every lead sold). Secondary metrics: 'Good' % (Closed + EP stages) and 'Bad' %.

Data checks and decisions

Check	Finding	Decision
Blank CallStatus	2,140 leads (70.8%); share stable at 63-80% every month	Treated as Unknown (neither good nor bad); not removed
Widget names	14 raw names; 300250 and 302252 are the same ad (per brief)	Merged to 11 designs; size and form pages parsed out
Partner spelling	'google' and 'Google' are separate values	Lower-cased and merged; 'advertise.com' (3) combined with AdKnowledge
Call-centre leads	271 leads (9%): own zone, own debt bucket ($7.5-15k), no campaign fields	Kept; shown as a separate channel
Score fields	PhoneScore exists for 16% of June and ~99% from July; AddressScore from July	Score analysis limited to Jul-Sep (n = 1,314)
IDs / e-mails	8 missing VendorLeadID, 1 duplicate, 133 repeated e-mails	Row-level analysis; no rows dropped
Status coding drift	'Doesn't Qualify' is ~0% in Apr-May but 11%-14% from June	Closed % used as the anchor metric; 'Bad' trend not over-interpreted
Empty fields	IP address column is empty; Sheet2 and Sheet3 are empty	Ignored (as the brief says)
Student-loan flag	Derived from AdGroup, Keyword and SearchQuery text ('student', 'loan default', 'default help'): 477 leads	Used as a segment (Google 414, Yahoo 62, other 1)

Statistical approach

- **Trend (Q1):** Cochran-Armitage and logistic-regression trend tests on Closed vs time, plus an early/late split (chi-square). Robustness: excluding September, online leads only, excluding student-loan traffic, the Good and Bad metrics, and a mix-adjusted model.
- **Drivers (Q2):** chi-square test per dimension (strength via Cramér's V), each segment vs the rest of the data (Fisher / chi-square), Benjamini-Hochberg false-discovery correction across all ~90 segments, and a multivariate logistic regression on online leads to separate overlapping factors.
- **Plan (Q3):** scenario arithmetic on observed data, a time-split check (rules examined on Apr-Jun, evaluated on Jul-Sep), and a break-even model for the CPL deal.
- **Caveat on scale:** with 245 closes in total, many segments are small; intervals are wide and I flag results that are suggestive rather than proven.

2. Q1 - Are lead-quality trends significant?

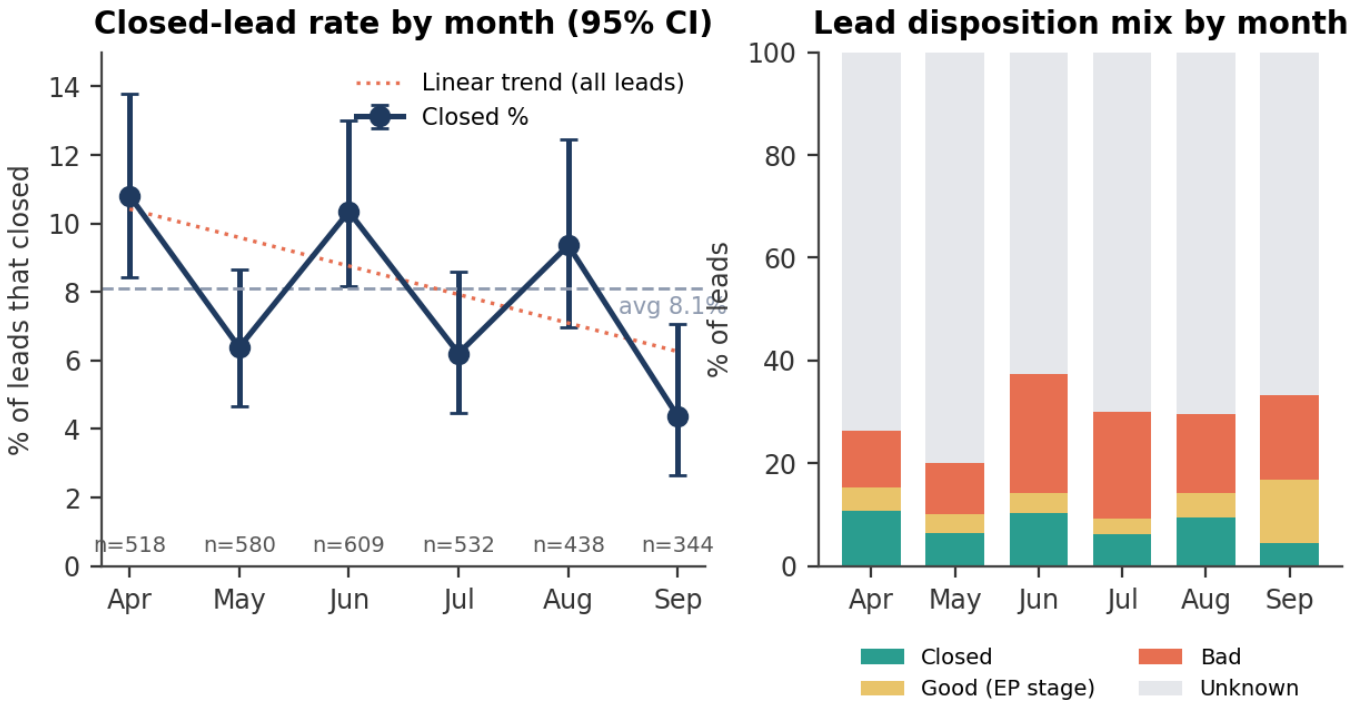


Figure 1. Closed rate fell in a noisy, stair-step pattern (left). The right panel shows the disposition mix; note September's large 'EP stage' share (yellow) that has not yet closed.

Test	n	Result	p-value	Interpretation
All leads, Closed	3,021	OR 0.89 per 30 days (CI 0.82-0.97)	0.006	Significant decline
Apr-Jun vs Jul-Sep	3,021	9.1% vs 6.8%	0.018	Significant decline
Mix-adjusted (online leads)	2,750	OR 0.85 per 30 days (CI 0.77-0.94)	0.001	Not explained by traffic mix, debt level or form
Online-form leads only	2,750	OR 0.87 per 30 days (CI 0.80-0.95)	0.001	Same direction
Excluding student-loan traffic	2,544	OR 0.89 per 30 days (CI 0.82-0.96)	0.004	Same direction
Excluding September	2,677	OR 0.94 per 30 days (CI 0.85-1.03)	0.18	**Not significant**: result hinges on September
Closed + EP-stage ('good')	3,021	OR 1.00 per 30 days (CI 0.94-1.07)	0.88	**Flat**: pipeline entering the close stage is stable
Bad share (see note)	3,021	OR 1.11 per 30 days (CI 1.04-1.17)	<0.001	Rises, but coding changed (see text)

Table 1. Trend tests (OR = odds ratio per 30 days for a lead closing; OR below 1 means falling quality). Cochran-Armitage on monthly counts gives the same conclusion for all leads (p = 0.021).

What this means

- **Direction and significance.** Closed rate declined from 10.8% in April to 4.4% in September. Both the early/late split and the trend regression are significant at 5%.
- **Why I would not call it settled.** (1) Excluding September removes significance. (2) September's Closed rate is lowest while its Closed + EP-stage rate is the *highest* of any month (16.9%); the 'EP Confirmed' stage is 10.8% of September leads vs about 3-5% before. These leads look like they are still moving through the funnel. The data has no close date, so maturation cannot be separated from a true decline.
- **The 'Bad' share is not a clean trend.** It looks significantly rising, but 'Doesn't Qualify' is almost absent in April-May and appears at 8-14% from June, which suggests the advertiser began recording that status around June. I therefore anchor on Closed %.
- **Traffic mix explains the July dip, not the whole story.** Student-loan traffic (which closes at ~1%) rose from 12% of leads in April to 29% in July, then fell to 5% in August (Figure 2). Excluding it lifts July from 6.2% to 8.8%, but September stays low (4.5%).

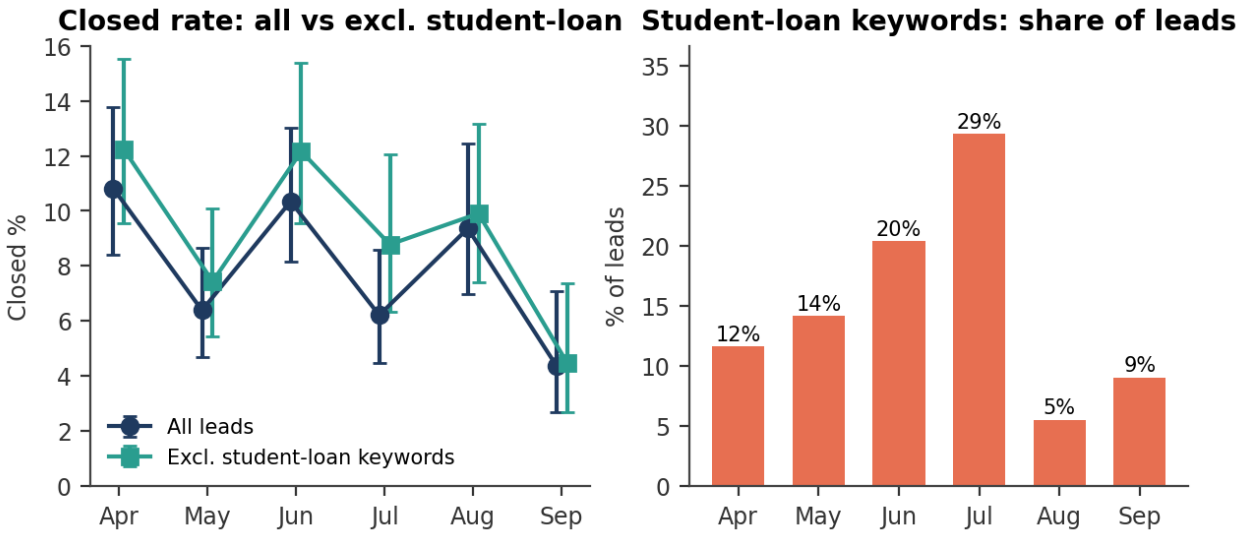


Figure 2. Student-loan keyword traffic peaked in June-July; removing it lifts every month but does not remove the September dip.

Recommendation: request close dates (or re-pull outcomes 30-60 days later) and track Closed % by lead-creation cohort, comparing cohorts at the same age. Until then, report the decline as 'significant but possibly overstated'.

3. Q2 - What drives lead quality?

I tested twelve dimensions covering where the ad ran, who filled it in and what they saw. Two stand out; the rest are not statistically distinguishable from noise at this sample size.

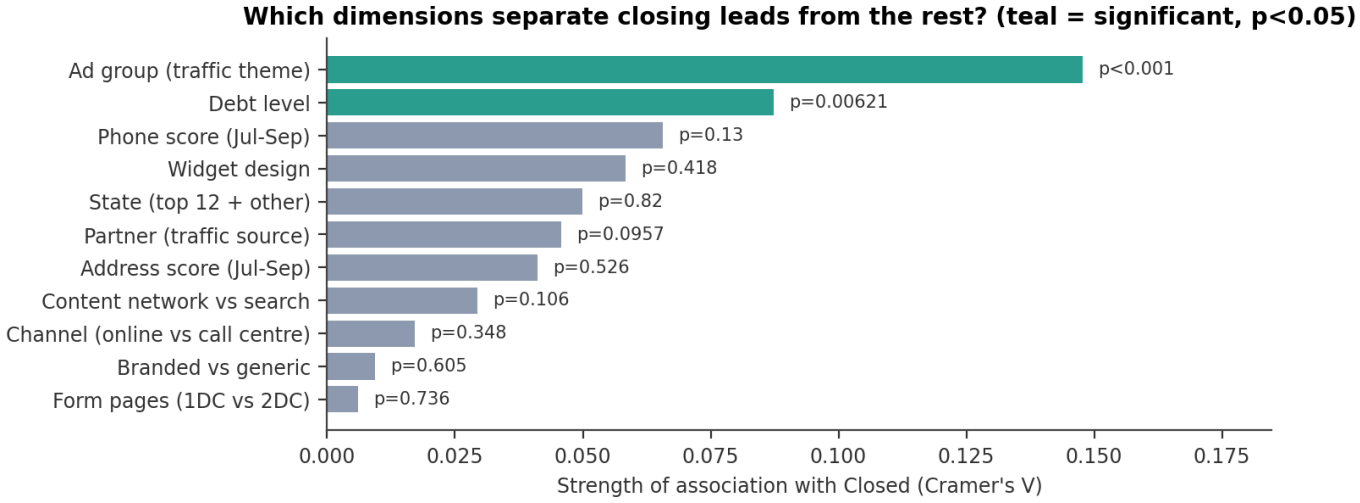


Figure 3. Strength of association between each dimension and a lead closing.

3.1 Traffic theme (ad group and keywords): the dominant driver

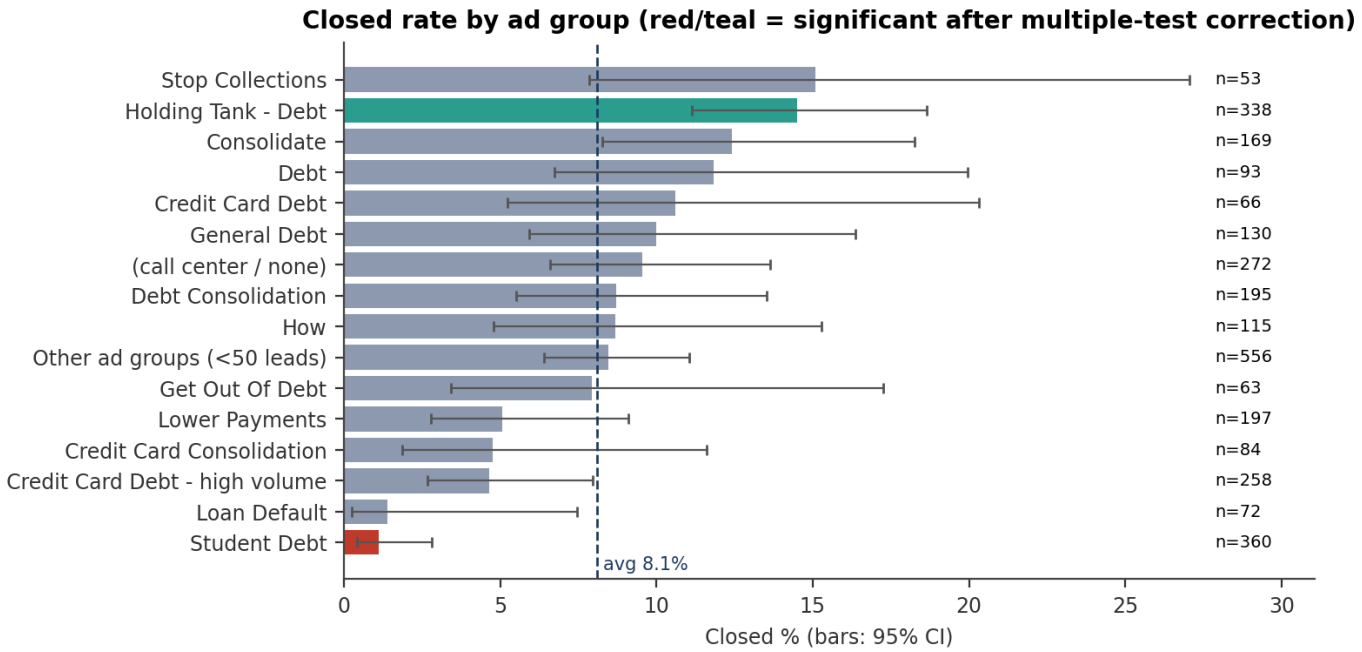


Figure 4. Closed rate by ad group. Red = significantly below average, teal = significantly above, after false-discovery correction.

- **Student-loan / loan-default traffic is almost worthless to the advertiser.** The 'Student Debt' ad group closed 4 of 360 leads (1.1%). Across all ad groups and keywords that mention student loans or loan default: 5 closes from 477 leads (1.0%) vs 9.4% elsewhere (p <0.001). Their 'bad' share is also high (31-50% in June-September). A plausible reason is that this debt type is often not eligible for the advertiser's programme; the advertiser can confirm.
- **'Holding Tank - Debt' is the strongest group:** 14.5% closed on 338 leads, 1.8x the average. 'Stop Collections' (15%), 'Consolidate' (12.5%) and 'Debt' (11.8%) are also above average, but with 50-170 leads each they are suggestive, not significant after correction.
- **Watch-list:** 'Credit Card Debt - high volume' (4.7%, n=258), 'Lower Payments' (5.1%, n=197) and 'Credit Card Consolidation' (4.8%, n=84) are below average but not significant individually.

3.2 Consumer debt level

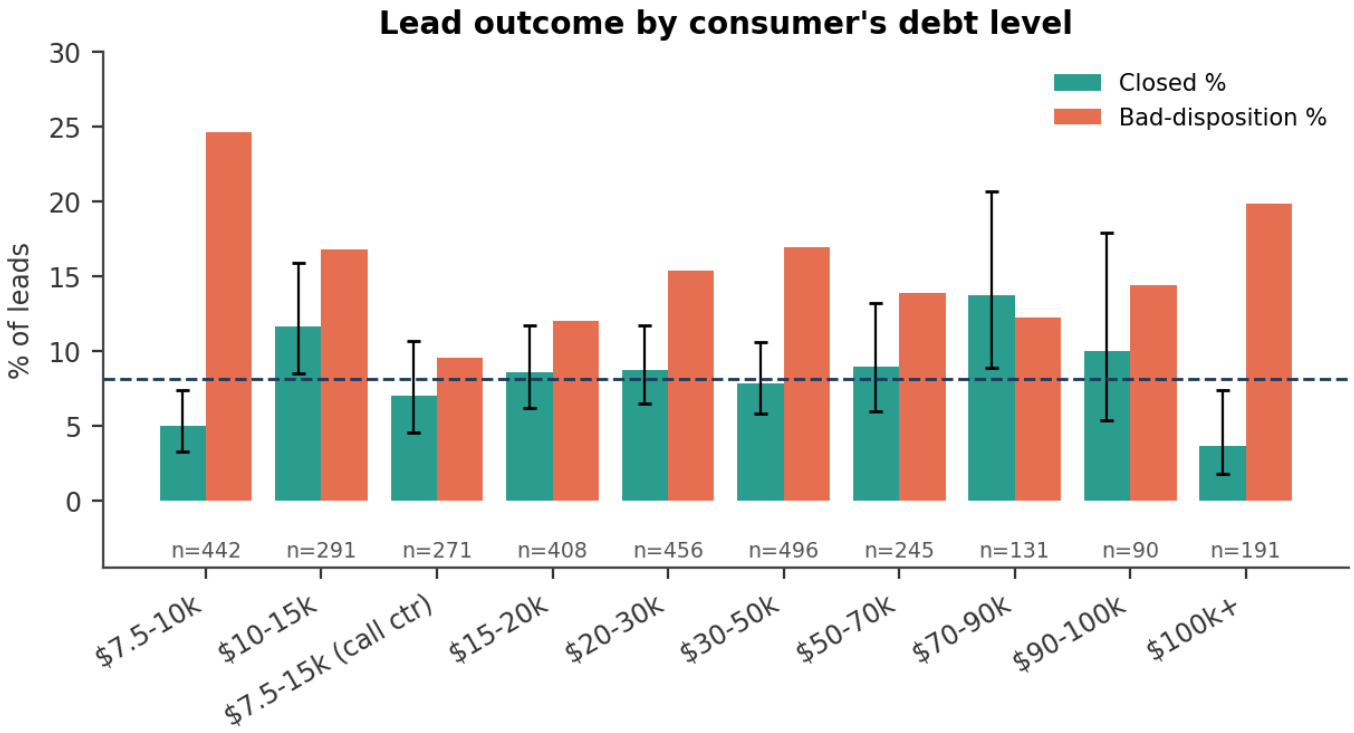


Figure 5. Closed % (teal, with 95% CI) and Bad-disposition % (orange) by debt level.

- Debt level is significantly associated with closing (p = 0.006). Leads with **$7.5-10k** of debt close at 5.0% and have the highest 'bad' share (24.7%), consistent with falling below the advertiser's minimum debt to qualify. Leads with **$100k+** close at 3.7% (19.9% bad), consistent with debt too large to settle or too little income.
- The best bands are **$70-90k** (13.7%, n=131) and **$10-15k** (11.7%). In the multivariate model (Figure 8) $70-100k is significantly better (OR 1.77, p = 0.020); $7.5-10k (OR 0.59, p = 0.060) and $100k+ (OR 0.47, p = 0.083) are suggestive.

3.3 Creative, form, partner and channel

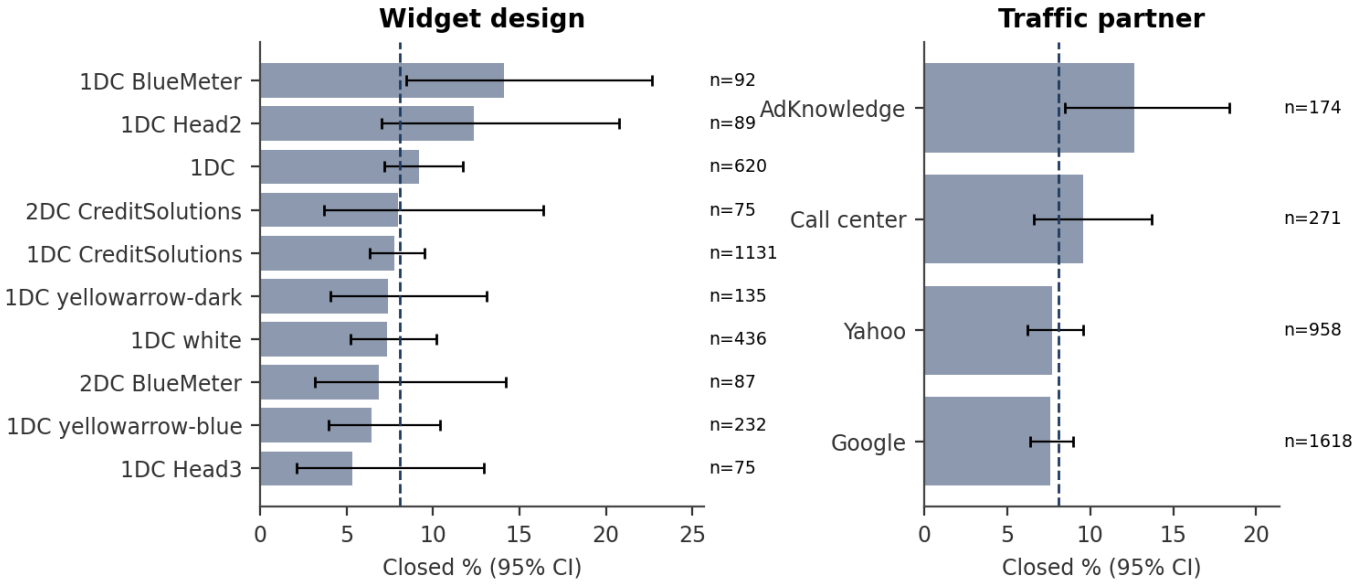


Figure 6. Closed % by widget design (n at least 70) and by traffic partner. Wide intervals overlap the average in almost every case.

- **Widget design does not matter reliably** (p = 0.42). 'BlueMeter' (14.1%, n=92) and 'Head2' (12.4%, n=89) look good but their intervals are very wide; they deserve a controlled test, not a rollout.

- **1-page vs 2-page form:** 8.1% vs 7.4% (p = 0.74); only 162 leads used 2 pages. **Branded vs generic ad:** 7.8% vs 8.3% (p = 0.60). Showing the advertiser's logo does not change lead quality.
- **Partner differences are mostly a traffic-mix effect.** AdKnowledge looks best (12.3%) and Google/Yahoo are about 7.7% (raw p = 0.096), but after controlling for ad group the partner effects are not significant (Figure 8). Content-network traffic closes at 5.7% vs 8.4% (p = 0.11), also not significant.
- **Call-centre leads** close at 9.6% vs 8.0% online (p = 0.35, not different), but have a markedly higher 'bad' share (24.4% vs 15.3%, p = <0.001), pointing to data-entry or screening issues rather than weaker buyers.
- **State** is not significant (p = 0.82); the top 12 states range from 5.7% to 12.5% but with small counts.

3.4 Contact validation scores (Jul-Sep only)

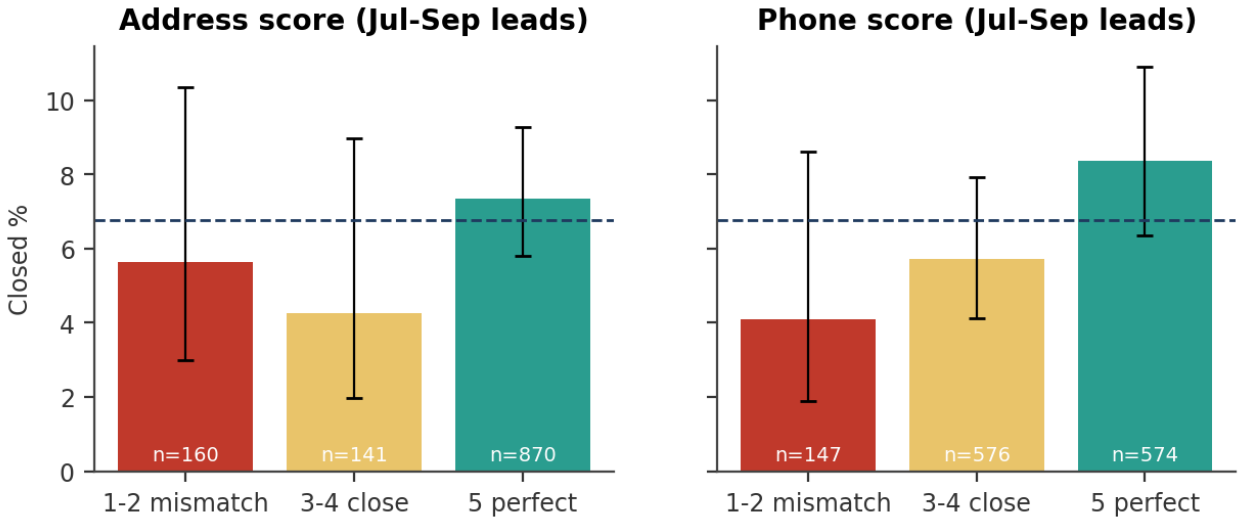


Figure 7. Closed % by validation score band (dashed line = Jul-Sep average of 6.8%).

- **PhoneScore:** leads with a perfect phone match close at 8.4% vs 5.4% for all other scores (p = 0.034 for this single comparison; the four-band chi-square across all levels is not significant, p = 0.13). Treat as a promising pre-filter to pilot.
- **AddressScore** shows a weaker, non-significant pattern (p = 0.53). Scores were only available for part of the period, which limits power.

3.5 All drivers together

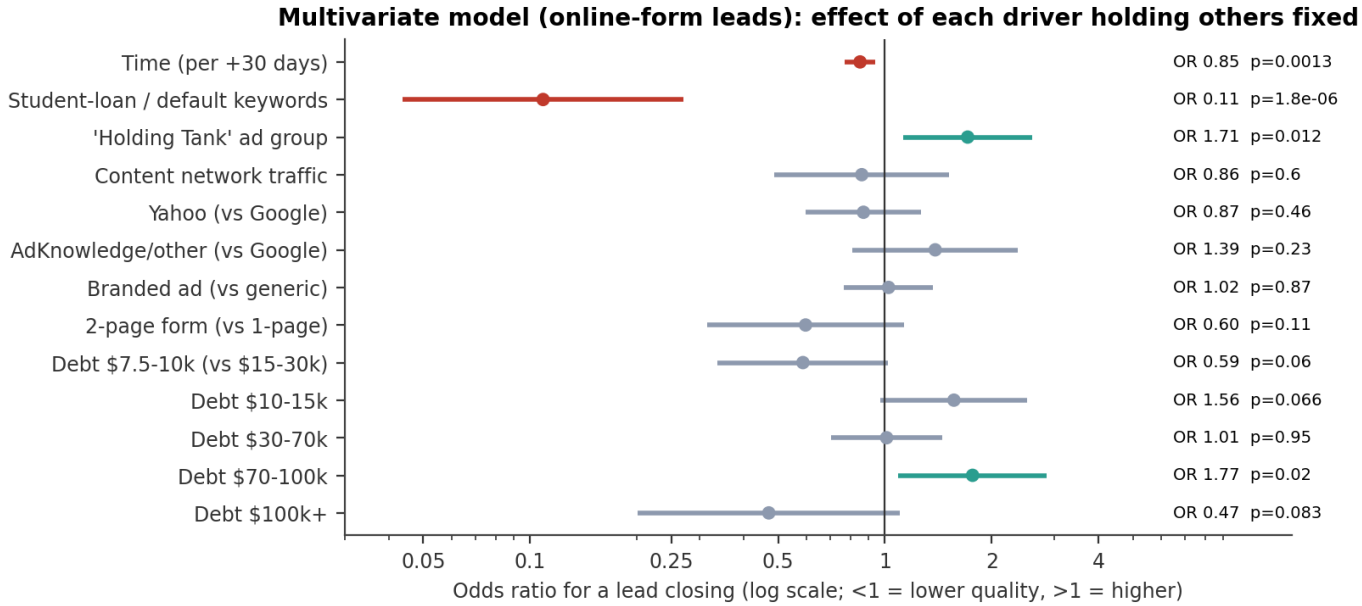


Figure 8. Logistic regression on 2,750 online-form leads. Odds ratio above 1 = more likely to close. Teal/red = significant at 5%.

- With everything else held fixed, only three things stand out: **student-loan keywords** (OR 0.11: about 89% lower odds of closing, p <0.001), the **'Holding Tank'** group (OR 1.71, p = 0.012) and the **$70-100k** debt band (OR 1.77, p = 0.020). Partner, content network, branded ad and form length add nothing once these are included.
- Time remains significantly negative (OR 0.85 per 30 days, p = 0.001) even after adjustment, so the decline in Q1 is not simply a change in the traffic mix; it may reflect September maturation or a genuine fall.

4. Q3 - Can we deliver a 20% quality lift?

The ask: raise Closed % from 8.0% to 9.6%. On this dataset's 3,021 leads that is about 45 more closes at the same volume (290 vs 245). **The economics:** a +20% CPL ($30 to $33) is worth more than the lead volume given up as long as volume stays above **83.3%** of today's (1 / 1.2). That break-even is the constraint on how aggressively we can filter.

4.1 Options that only remove traffic

Option	Volume kept	Closed % (95% CI)	Lift vs 8.1%	Apr-Jun / Jul-Sep rate	Hits 9.6%?	Revenue index*
Today	100%	8.1% (7.2%-9.1%)	-	9.1% / 6.8%	-	1.00
A. Stop student-loan / loan-default traffic	84%	9.4% (8.4%-10.6%)	+16%	10.5% / 8.0%	No (short by 0.17 pts)	0.84
B. A + exclude $100k+ debt	80%	9.7% (8.6%-10.9%)	+19%	10.8% / 8.2%	Yes	0.96
C. B + exclude $7.5-10k debt	67%	10.5% (9.2%-11.9%)	+30%	11.3% / 9.3%	Yes	0.80

*Table 2. *Revenue index = volume kept x 1.2 if the 9.6% target is met, otherwise volume kept (CPL unchanged); today = 1.00. Gross publisher revenue only; traffic cost is not in the data (see limitations). 'Apr-Jun / Jul-Sep' shows the same rule applied to each half of the data.*

- **Option A is the clear first move.** It lifts quality by 16% while keeping 84% of volume, and it generalises: student-loan traffic closed 1.5% in Apr-Jun and 0.5% in Jul-Sep, and removing it raised the Jul-Sep rate from 6.8% to 8.0%. It is just short of 9.6% on its own (9.4%; the confidence interval includes 9.6%).
- **B reaches the target** (9.7%) but keeps only 80% of volume, below the 83.3% break-even, so gross revenue would be about 4% lower than today unless the traffic cost saved exceeds that.
- **C overshoots quality** (10.5%) but cuts volume by a third (revenue index 0.80); I would not recommend it. Note that the Jul-Sep volumes differ by rule, so the table gives an indication only.

4.2 The better route: replace removed traffic with better traffic

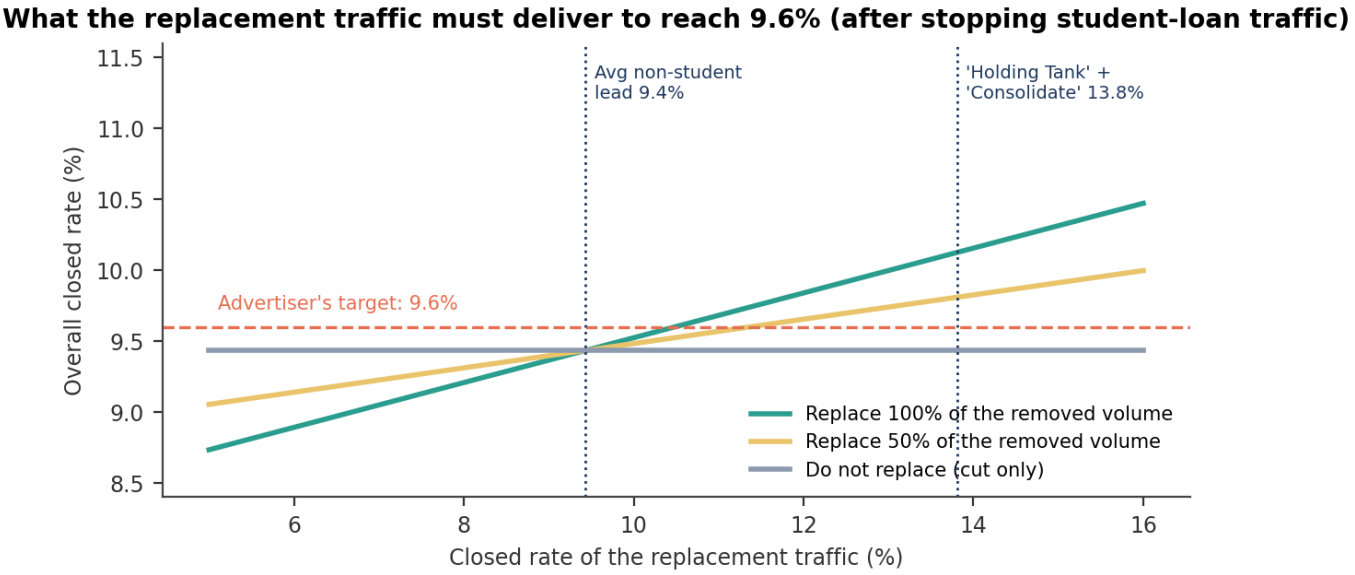


Figure 9. After stopping student-loan traffic (477 leads freed), the overall closed rate depends on how well the replacement traffic converts.

Replace this much of the freed volume	Replacement closes at 9.4% (average non-student)	10.5%	12.0%	13.8% (Holding Tank + Consolidate)
0% (cut only)	9.4%	9.4%	9.4%	9.4%
50% (volume 92%)	9.4%	9.5%	9.7% ✓	9.8% ✓
100% (volume 100%)	9.4%	9.6% ✓	9.8% ✓	10.1% ✓

Table 3. Overall closed rate by scenario; ✓ = reaches 9.6%. Revenue index is 1.20 for full replacement and 1.11 for 50% replacement when the target is met.

- To hit 9.6% at full volume, replacement traffic needs to close at **10.5% or better**. That is only about one point above the average non-student lead (9.4%) and well below the best ad groups (13.8% for 'Holding Tank' + 'Consolidate', n = 507). The risk is diminishing returns: scaling a keyword group usually lowers its marginal conversion.
- If replacement can be only half of the freed volume, it must close at about 12%, still within the range seen in these ad groups.
- Combining the debt screen with replacement gives extra headroom: at full replacement the overall rate would exceed 10%.

4.3 Recommended actions

#	Action	Evidence	Expected effect	Confidence
1	Add negative keywords and pause the 'Student Debt', 'Student Debt - high volume' and 'Loan Default' ad groups on Google and Yahoo	5 closes from 477 leads (p <0.001); holds in both halves of the period	+1.3 pts: 8.1% to 9.4%; -16% volume	High
2	Redirect the freed budget into 'Holding Tank - Debt', 'Consolidate', 'Stop Collections' and 'Debt'; scale in steps and monitor marginal close rate	12-15% closed; Holding Tank significant after correction	+0.2-0.7 pts depending on marginal conversion; restores volume	Medium
3	Qualify debt level up front (collect first, show clear minimum-debt messaging; route or reprice $100k+ and under $10k)	$7.5-10k: 5.0% with 25% bad; $100k+: 3.7%	+0.2 pts (excluding $100k+) for about 5% of volume	Medium
4	Pilot a phone-validation gate or lower bid for leads with imperfect PhoneScore	8.4% vs 5.4% (p = 0.034, Jul-Sep only)	Unknown until piloted (perfect-match leads close about 3 pts higher)	Low-Medium
5	A/B test 'BlueMeter' and 'Head2' creatives against the control	14.1% and 12.4% but n of about 90 each	Unknown; roll out only if the test confirms	Low
6	Review call-centre data entry and screening	24.4% bad vs 15.3% online (p = <0.001)	Fewer wasted advertiser calls; protects CPL deal	Medium
7	Agree the measurement method with the advertiser before the CPL change	See below	Avoids disputes caused by noise	-

Measuring success

- Detecting a lift from 8.0% to 9.6% with 80% power needs about **4,912 leads per arm**. At roughly 500 leads a month split across two arms, a test on this publisher's data alone would take well over a year, so changes should be evaluated on pooled network volume or with a sequential before/after comparison.
- Proposing Closed + EP-stage as a leading indicator (13% base rate) needs only about 2,840 leads per arm for a 20% relative lift and shows results sooner, because it does not wait for the close.
- Compare cohorts of equal age, because recent leads are always still maturing.

5. Limitations and next steps

- **No close dates or cost data:** maturation cannot be isolated and net margin cannot be computed. The revenue index covers gross CPL revenue only; cutting weak traffic would also save media cost, which would improve the case for options A to C.
- **Selection risk:** segments were examined after seeing the data. I corrected for multiple testing and checked the student-loan rule across time halves. An automated search for cuts on Apr-Jun chose 'Student Debt' first and then two small widget designs; those design cuts added nothing in Jul-Sep (7.6% vs 8.0% for the student-loan rule alone), consistent with design differences being noise.
- **Blank CallStatus (71%):** assumed Unknown. If many blanks are simply un-worked leads, true Closed % would be higher than 8.1% and trends could shift.
- **Small segments:** many ad groups, designs and states have fewer than 100 leads, so only large effects are detectable. 'Suggestive' findings should be tested, not acted on at scale.

- **Next steps:** (1) get close dates, (2) launch action 1 immediately, (3) run the replacement and debt-screen pilots with a pre-agreed evaluation window, (4) re-run this analysis monthly on mature cohorts.

Appendix A. Segment results

Every segment with at least 70 leads from the ad-group, debt, partner, design, form-page, branded and content-network dimensions, sorted by Closed rate. 'vs rest' is a two-sided test of the segment against all other leads; q is the Benjamini-Hochberg adjusted value across all segments tested.

Dimension	Segment	n	Closed	Rate	95% CI	p vs rest	q
Ad group	Holding Tank - Debt	338	49	14.5%	11.1%-18.6%	<0.001	<0.001
Widget design	1DC-BlueMeter	92	13	14.1%	8.4%-22.7%	0.032	0.22
Debt level	$70-90k	131	18	13.7%	8.9%-20.7%	0.016	0.22
Partner	adknowledge/other	174	22	12.6%	8.5%-18.4%	0.024	0.22
Ad group	Consolidate	169	21	12.4%	8.3%-18.2%	0.034	0.22
Widget design	1DC-Head2	89	11	12.4%	7.0%-20.8%	0.14	0.53
Ad group	Debt	93	11	11.8%	6.7%-20.0%	0.18	0.58
Debt level	$10-15k	291	34	11.7%	8.5%-15.9%	0.019	0.22
Debt level	$90-100k	90	9	10.0%	5.4%-17.9%	0.50	0.82
Ad group	General Debt	130	13	10.0%	5.9%-16.4%	0.42	0.82
Channel	Call center	271	26	9.6%	6.6%-13.7%	0.35	0.74
Partner	call_center	271	26	9.6%	6.6%-13.7%	0.35	0.74
Ad group	(call center / none)	272	26	9.6%	6.6%-13.6%	0.36	0.74
Widget design	1DC	620	57	9.2%	7.2%-11.7%	0.27	0.71
Debt level	$50-70k	245	22	9.0%	6.0%-13.2%	0.60	0.82
Debt level	$20-30k	456	40	8.8%	6.5%-11.7%	0.57	0.82
Ad group	Debt Consolidation	195	17	8.7%	5.5%-13.5%	0.75	0.88
Ad group	How	115	10	8.7%	4.8%-15.3%	0.81	0.88
Debt level	$15-20k	408	35	8.6%	6.2%-11.7%	0.71	0.88
Content network vs search	Search / other	2,722	228	8.4%	7.4%-9.5%	0.11	0.45
Branded vs generic	Generic	1,815	151	8.3%	7.1%-9.7%	0.60	0.82
Form pages	1DC	2,859	233	8.1%	7.2%-9.2%	0.74	0.88
Widget design	2DC-CreditSolutions	75	6	8.0%	3.7%-16.4%	0.97	0.97
Channel	Online form	2,750	219	8.0%	7.0%-9.0%	0.35	0.74
Debt level	$30-50k	496	39	7.9%	5.8%-10.6%	0.83	0.88
Branded vs generic	Branded (logo)	1,206	94	7.8%	6.4%-9.4%	0.60	0.82
Widget design	1DC-CreditSolutions	1,131	88	7.8%	6.4%-9.5%	0.61	0.82
Partner	yahoo	958	74	7.7%	6.2%-9.6%	0.60	0.82
Partner	google	1,618	123	7.6%	6.4%-9.0%	0.27	0.71
Widget design	1DC-yellowarrow-dark	135	10	7.4%	4.1%-13.1%	0.76	0.88
Form pages	2DC	162	12	7.4%	4.3%-12.5%	0.74	0.88
Widget design	1DC-white	436	32	7.3%	5.2%-10.2%	0.52	0.82
Debt level	$7.5-15k (call ctr)	271	19	7.0%	4.5%-10.7%	0.49	0.82
Widget design	2DC-BlueMeter	87	6	6.9%	3.2%-14.2%	0.67	0.88
Widget design	1DC-yellowarrow-blue	232	15	6.5%	4.0%-10.4%	0.34	0.74
Content network vs search	Content network	299	17	5.7%	3.6%-8.9%	0.11	0.45
Widget design	1DC-Head3	75	4	5.3%	2.1%-12.9%	0.52	0.82

Dimension	Segment	n	Closed	Rate	95% CI	p vs rest	q
Ad group	Lower Payments	197	10	5.1%	2.8%-9.1%	0.11	0.45
Debt level	$7.5-10k	442	22	5.0%	3.3%-7.4%	0.009	0.21
Ad group	Credit Card Consolidation	84	4	4.8%	1.9%-11.6%	0.31	0.74
Ad group	Credit Card Debt - high volume	258	12	4.7%	2.7%-8.0%	0.033	0.22
Debt level	$100k+	191	7	3.7%	1.8%-7.4%	0.020	0.22
Ad group	Loan Default	72	1	1.4%	0.2%-7.5%	0.028	0.22
Ad group	Student Debt	360	4	1.1%	0.4%-2.8%	<0.001	<0.001

Appendix B. Reproducibility

All numbers in this report were produced programmatically from the supplied workbook (Python: pandas, SciPy, matplotlib). Logistic regressions were fitted by iteratively re-weighted least squares with Wald confidence intervals. Closed % uses all 3,021 leads as the denominator. The student-loan flag is a keyword match on ad group, keyword and search-query text; other definitions are in Section 1.